

Extension to Pension Run

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September 24, 2026

1. Introduction

Why do retirees give up longevity insurance? In the canonical life-cycle model, an individual who faces an uncertain lifetime should convert a substantial share of wealth into an annuity. Pooling mortality risk allows an annuity to sustain consumption for as long as the retiree lives, a protection that self-insurance cannot provide as cheaply ([Davidoff, Brown, & Diamond, 2005](#); [Yaari, 1965](#)). Yet observed annuitization is limited: even when retirees are offered a direct choice between a lump-sum benefit and a stream of lifetime payments, many choose the lump sum. This gap between the theoretical value of longevity insurance and actual choices is the annuitization puzzle ([Benartzi, Previtro, & Thaler, 2011](#); [Brown, Kling, Mullainathan, & Wrobel, 2008](#)).

Most explanations of the puzzle treat the value of the promised annuity as independent of other retirees' choices. That assumption is natural for an actuarially fair contract backed by a solvent insurer, but it is less innocuous for benefits paid by a financially vulnerable pension fund. A lifetime payment is valuable only while the provider remains able and willing to pay. When a fund is expected to become depleted, the effective maturity of its monthly benefit is shortened. A retiree may then rationally prefer an immediate lump sum even when the scheduled, undiscounted value of monthly benefits is larger. Low annuitization can therefore reflect not only preferences for liquidity or bequests, but also exposure to provider-solvency risk.

The central tension is that the annuity is the more generous option if all promised payments are honored, yet its generosity can undermine the promise itself. When early cohorts choose monthly benefits, they build a persistent stock of long-lived liabilities. In the environment we study, those obligations eventually deplete the fund, whereas the fund remains sustainable if every generation instead takes the less generous lump-sum benefit. Insolvency is therefore not simply an exogenous reason to avoid annuitization; it can be the equilibrium consequence of earlier cohorts accepting the annuity offer.

This intergenerational feedback gives the pension run two stages. Early recipients choose the more valuable monthly benefit and gradually weaken the fund's solvency. Once the accumulated obligations make depletion sufficiently likely, later retirees discount the promised

stream and switch toward immediate lump-sum claims. These large current withdrawals can bring depletion forward and further reduce the value of annuitization. We call this feedback a *pension run*. Unlike a standard bank run, retirees do not withdraw identical demandable claims at the same time. They make a one-time choice as they retire, so aggregate fragility develops through a sequence of benefit-maturity decisions across generations.

This paper develops a continuous-time model of that mechanism. New retirees choose between an immediate lump-sum payment and a life-contingent monthly benefit. Their choice depends on the present value of monthly payments, which in turn depends on the stochastic time at which the pension fund is depleted. Individual choices determine both current withdrawals and the stock of monthly beneficiaries, feeding back into the fund's dynamics and its bankruptcy time. The equilibrium is summarized by a nonlinear valuation equation linking the fund balance, outstanding pension liabilities, annuity demand, and solvency. The framework thus turns one possible explanation of the annuitization puzzle into an equilibrium source of pension-system fragility.

Related literature. This paper contributes first to the literature on annuity demand. The benchmark result is that households facing uncertain lifetimes value annuities because they insure consumption against longevity risk (Davidoff et al., 2005; Yaari, 1965). Observed behavior falls well short of this benchmark. Using U.S. Health and Retirement Study data, Brown (2001) finds that the life-cycle value of an annuity predicts whether defined-contribution balances are annuitized, but leaves much of the observed choice unexplained. Dushi and Webb (2004) likewise document very low purchases of additional annuities among older U.S. households. Evidence from the United Kingdom also shows limited participation in the voluntary annuity market (Inkmann, Lopes, & Michaelides, 2011). Even when an annuity is offered directly within a pension plan, a substantial minority takes cash: in U.S. pension data, Hurd and Panis (2006) find that 20 percent of benefits offering a lump-sum option are cashed out. Taken together, these studies establish a persistent gap between the large theoretical gains from longevity insurance and retirees' willingness to annuitize, although its magnitude varies with the institutional setting (Benartzi et al., 2011).

The literature offers several explanations for this gap. Adverse selection raises the price and lowers the money's worth of voluntary annuities because individuals with private information that they are long-lived are more likely to purchase them (Finkelstein & Poterba, 2002). Fixed nominal annuities expose retirees to inflation risk, while inflation-indexed products have historically been scarce in private markets (?). Public pensions and defined-benefit plans also pre-annuitize a large share of household wealth, reducing the marginal value of purchasing additional longevity insurance (Dushi & Webb, 2004). Health and medical-expense shocks create a separate demand for liquid wealth. When high medical costs are correlated with changes in mortality, the remaining value of an annuity can be especially low precisely when those expenses arise (Reichling & Smetters, 2015). Bequest motives, other liquidity needs, and the framing of annuity products provide additional explanations

(Brown et al., 2008).

Most closely related to our mechanism, Li, Neumuller, and Rothschild (2021) show that real and perceived provider-insolvency risk, together with imperfect information about guaranty protection, can substantially reduce annuity demand and welfare. In their analysis, households take the provider’s insolvency risk as given when choosing how much to annuitize. In our setting, insolvency is endogenous: retirees’ choices between lump-sum and monthly benefits alter the fund’s cash flows and hence its probability of depletion. Annuity demand, provider solvency, and the value of the annuity are therefore jointly determined in equilibrium.

Second, the paper relates to research on the sustainability and reform of public pensions. Demographic aging erodes the contribution base and expands benefit obligations, creating adjustments in taxes, benefits, retirement ages, labor supply, and private saving (Attanasio, Kitao, & Violante, 2007; Börsch-Supan, Härtl, & Leite, 2016). Structural analyses quantify how pension institutions and reforms reshape retirement and life-cycle decisions (Daminato & Padula, 2024; Delalibera, Ferreira, & Machado Parente, 2026; French & Jones, 2011; Kitao, 2018). Our focus is complementary. We hold the principal policy parameters fixed and ask how a benefit-withdrawal option can propagate financial stress even before a reform occurs. This introduces beneficiary composition—the division of new retirees between immediate and deferred claims—as an endogenous determinant of pension solvency.

Relatedly, our mechanism connects to evidence that retirement decisions respond to expectations about future pension policy. Bi and Zubairy (2023) show that the timing of pension retrenchments matters: immediately implemented reforms delay retirement, whereas reforms announced with long implementation lags can bring retirement forward, particularly when the reform is fundamental or trust in government is low. We likewise emphasize forward-looking responses to the credibility of future benefits. In our setting, however, the relevant uncertainty arises endogenously from the fund’s financial state, and retirees’ responses feed back into that state rather than merely responding to an externally specified reform path.

Finally, the paper is related to models of runs and maturity choice. The classic bank-run literature shows how individually optimal withdrawals can generate aggregate fragility (Diamond & Dybvig, 1983; Goldstein & Pauzner, 2005). The closest structural analogue is the dynamic debt-run model of He and Xiong (2012), in which creditors arrive sequentially and decide whether to roll over maturing short-term debt. We retain the stochastic fundamental, staggered decisions, and endogenous default time, but the underlying claim is different. A retiree chooses once between an immediate payment and a life-contingent stream; this choice changes both current outflows and the evolving stock of monthly-benefit recipients. The pension run therefore operates through endogenous benefit maturity and demographic turnover, rather than repeated rollover of fixed-maturity debt.

2. Institutional Background

2.1. Labor Insurance old-age benefits

The empirical setting is the old-age benefit provided by Taiwan’s Labor Insurance program. This social-insurance benefit is distinct from the Labor Pension system, under which employers contribute to workers’ individual retirement accounts. Labor Insurance is financed collectively by premiums and the assets of the Labor Insurance Fund, and benefits are administered by the Bureau of Labor Insurance (BLI).

An old-age pension was introduced on January 1, 2009. An insured worker with at least 15 years of coverage may claim a monthly pension after leaving employment and reaching the statutory claim age. The statutory age increased gradually from 60 and reaches 65 in 2026. A worker may claim up to five years early or defer for up to five years; the monthly benefit is reduced or increased by 4 percent for each year. At the normal claim age, the BLI pays the larger amount implied by

$$p = \max \{0.00775 wY + 3,000, 0.0155 wY\}, \quad (1)$$

where w is the average of the worker’s highest 60 monthly insured salaries and Y is years of coverage.

The 2009 reform created the choice studied in this paper. Workers with any Labor Insurance coverage before December 31, 2008 may choose either the monthly pension or the pre-reform *one-time old-age benefit*; the choice is irreversible once the BLI approves and pays the claim. Workers first insured in 2009 or later cannot choose the pre-reform benefit. The one-time benefit generally pays one month of average insured salary for each of the first 15 coverage years and two months for each additional year, subject to a ceiling of 45 months before age 60 and 50 months including post-60 coverage. This grandfathered payment is the lump sum in our model. It should not be confused with the separately defined *old-age lump-sum benefit* for workers with fewer than 15 years of coverage, whom we exclude from the choice sample.

The statutory formulas make the annuity potentially much more generous than the lump sum. For example, a worker with an average insured salary of NT\$45,800 and 30 years of coverage receives NT\$2.061 million under the one-time formula but NT\$21,297 per month under the second formula in [equation \(1\)](#). If the worker receives the pension for 12 years, its present value at a 2 percent annual discount rate is approximately NT\$2.72 million. The monthly option therefore offers valuable longevity insurance as well as a larger expected transfer, but it also creates a persistent liability for the common fund. This difference is the institutional basis for the tradeoff in our model.

Taiwan’s government has made direct budgetary contributions to the Labor Insurance Fund since 2020. Accordingly, “bankruptcy” in the model does not mean that current law

mechanically terminates benefits when recorded assets reach zero. It denotes exhaustion of the fund under existing contribution and benefit rules—the point at which continued payment requires an additional fiscal transfer or a reform. The baseline estimation ends in December 2019 to identify the fund’s endogenous dynamics before regular government injections; the later observations remain useful for describing choices and validating the entry reconstruction.

3. Data

3.1. Sources and sample construction

We construct a monthly panel beginning in January 2009, when the old-age pension was introduced. The current data vintage ends in June 2026, the latest month for which all component series are available. The panel combines three official sources. First, monthly reports from the Bureau of Labor Funds (BLF) provide the exact month-end balance and investment return of the Labor Insurance Fund. We read the balance from the report’s asset-allocation table rather than its rounded narrative total. The fund included both ordinary and occupational-accident insurance through April 2022; following the institutional separation in May 2022, the reported Labor Insurance Fund contains only ordinary insurance. Second, BLI statistical tables provide the stock of old-age-pension recipients, pension recipients receiving their first payment, and claims for the one-time old-age benefit. Third, Ministry of the Interior (MOI) population files provide monthly registered population and annual population and deaths by age.

We retain only one-time old-age benefit claims as lump-sum choices. In particular, we exclude the old-age lump-sum benefit for short coverage histories and the old-age difference payment because neither represents the same choice between the grandfathered lump sum and the monthly pension. Let L_t denote the number of qualifying one-time claims in month t , Q_t the stock of monthly pension recipients, and A_t the number entering the monthly scheme. The empirical monthly-choice share is

$$q_t = \frac{A_t}{A_t + L_t}. \quad (2)$$

The two state variables used in estimation are the fund balance per registered resident, B_t/N_t , and monthly pension recipients per resident, Q_t/N_t .

3.2. Reconstructing entries into the monthly scheme

The BLI reports Q_t throughout the sample, but identifies first-payment recipients only from May 2022 onward. A change in Q_t is not itself an entry measure because recipients leave the stock when they die. For the earlier period, we reconstruct monthly entries from the

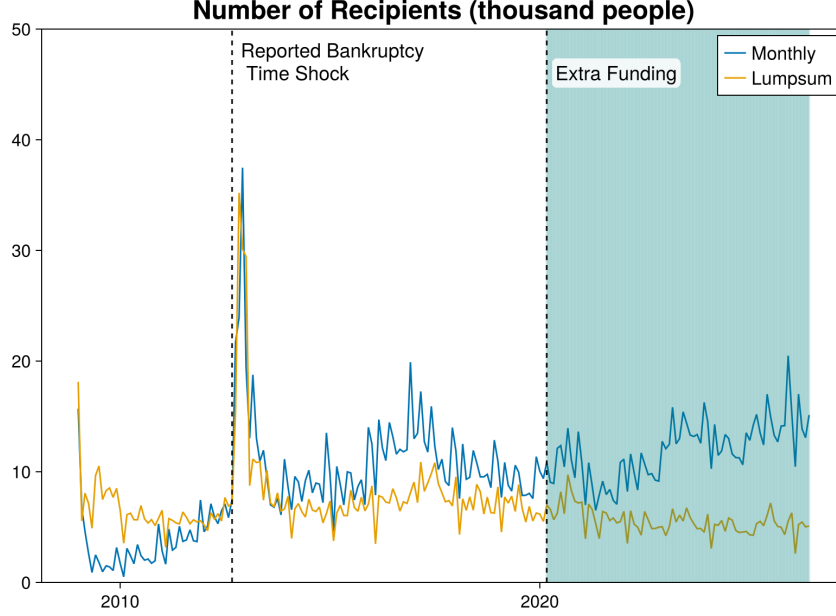


Figure 1: New Labor Insurance old-age benefit recipients

accounting identity

$$\hat{A}_t = \max \{0, Q_t - Q_{t-1} + \delta_t Q_{t-1} + c\}, \quad (3)$$

where δ_t is the monthly mortality probability for Taiwan residents aged 60 and older. We calculate the corresponding annual death probability from MOI age-specific deaths and population, then convert it to a monthly probability as $1 - (1 - d_t)^{1/12}$. If the current year's age table has not yet been released, we carry forward the latest available annual rate.

The constant c corrects for systematic differences between the stock-flow identity and reported first payments. We estimate it over the post-May-2022 overlap as the mean of $A_t - (Q_t - Q_{t-1} + \delta_t Q_{t-1})$. In the current data vintage, the overlap contains 50 months and yields $c = -2,452.84$ recipients per month; the mean absolute reconstruction error in that validation period is 179 recipients. We use reported first payments whenever they are available and the adjusted reconstruction beforehand. Equation (3) assumes that mortality is the only exit from the recipient stock and applies the mortality of all residents aged 60 and older to Labor Insurance beneficiaries. The reconstructed early series should therefore be interpreted as a measurement of annuity entry rather than an exact administrative count.

4. Model

4.1. Demographics

Let b be the fertility rate and $m(s)$ the mortality rate. We assume that the individuals cannot live longer than T and $n(t, T) = 0$. Let $n(t, s)$ be the unnormalized population density. The

population dynamic is given by

$$\frac{\partial n}{\partial t}(t, s) = -\frac{\partial n}{\partial s}(t, s) - m(s)n(t, s), \quad n(t, 0) = b \int_0^T n(t, s) ds. \quad (4)$$

Let $N_t = \int_0^T n(t, s) ds$ be the total population. Detrending the population equation by $\hat{n}_t = n_t/N_t$ gives

$$\frac{\partial \hat{n}}{\partial t}(t, s) = -\left(\frac{\dot{N}(t)}{N(t)} + m(s)\right)\hat{n}(t, s) - \frac{\partial \hat{n}}{\partial s}(t, s), \quad \hat{n}(t, 0) = b. \quad (5)$$

The population growth rate η_t is given by

$$\begin{aligned} \frac{\dot{N}_t}{N_t} &= \frac{\int_0^T \frac{\partial n}{\partial t} ds}{\int_0^T n(t, s) ds} = \frac{\int_0^T -\frac{\partial n}{\partial s}(t, s) - m(s)n(t, s) ds}{\int_0^T n(t, s) ds} \\ &= \frac{-(n(t, T) - n(t, 0)) - \int_0^T m(s)n(t, s) ds}{\int_0^T n(t, s) ds} = b - \int_0^T m(s)\hat{n}(t, s) ds \end{aligned} \quad (6)$$

In the steady state, the population grows at a constant rate $\eta_t = \frac{\dot{N}(t)}{N(t)}$. The steady state $\bar{n}$ is given by

$$\bar{n}(s) = b \exp\left(-\int_0^s \eta + m(x) dx\right). \quad (7)$$

The normalization condition $\int_0^T \bar{n}(s) ds = 1$ determines η :

$$b \int_0^T \exp\left(-\int_0^s \eta + m(x) dx\right) ds = 1. \quad (8)$$

4.2. Household

Retirees. Given the interest rate r , a retiree that dose not participate the monthly plan maximizes

$$V_t^\ell = \max \int_t^T e^{-\int_t^s \rho + m(x) dx} [u(c_s) + m(s)b(a_s)] ds \quad (9)$$

subject to

$$\dot{a}_t = r_t a_t + \text{tr}_t - c_t \quad (10)$$

$$a_t \geq 0. \quad (11)$$

The utility flow is given by

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \quad (12)$$

and the bequest motive follows [De Nardi \(2004\)](#) and [French and Jones \(2011\)](#):

$$b(a) = \theta_b \frac{(a + \kappa)^{1-\gamma}}{1 - \gamma}. \quad (13)$$

We assume that the transfer tr_t is a flow from the bequest, which is assumed to be distributed uniformly to all the agents alive.

For a retiree participating the monthly plan, the consumption policy solves

$$V_t^m = \max \int_t^T e^{-\int_t^s \rho + m(x) dx} [u(c_s) + m(s)b(a_s)] ds \quad (14)$$

subject to

$$\dot{a}_t = r_t a_t + \text{tr}_t + p - c_t \quad (15)$$

$$a_t \geq 0. \quad (16)$$

Workers. After aging T_w , the individuals can start to work. Given interest rate r_t and wage w_t , the households consume optimally to maximize

$$V_t^w = \mathbb{E}_t \left[\int_t^T e^{-\int_t^s \rho + m(x) dx} [u(c_s) - \psi(s) + m(s)b(a_s)] ds \right] \quad (17)$$

subject to

$$\dot{a}_t = r_t a_t + (y_t - \tau(y_t)) + \text{tr}_t - c_t \quad (18)$$

$$a_t \geq 0 \quad (19)$$

$$y_t = w_t \exp(\bar{z}(t) + z_t) \quad (20)$$

where $\bar{z}(t)$ is the earning profile and $z_t \in \mathcal{Z} = \{z_0, \dots, z_k\}$ follows a continuous time Markov chain with intensity matrix Λ . We take $z_0 = -\infty$ for the unemployment state. The initial z at T_w is drawn randomly from the invariant distribution associated with Λ . The ψ is the disutility to work. The pension tax τ is of the form

$$\tau(y) = \min \{\tau_0 y, \tau_1\} \quad (21)$$

which is truncated from above.

We allow the workers to retire at any age. Yet, the pension is only available for those retiring within the retirement window $[\underline{T}_r, \overline{T}_r]$. The retirement status

$$R_t \in \{\text{not to retire, retire with lumpsum, retire with pension}\}. \quad (22)$$

The retirees cannot return to work.

Youngs. The youngs are born with zero asset and cannot work before T_w . They are assumed to be hand to mouth so $c_t = tr_t$.

4.3. Firms

The production function is given by

$$Y_t = AK_t^\alpha L_t^{1-\alpha}. \quad (23)$$

Given r_t, w_t , the firms solve

$$\max_{K_t, L_t} AK_t^\alpha L_t^{1-\alpha} - r_t K_t - w_t L_t \quad (24)$$

where L_t is the aggregate effective labor. Taking the FOC and detrending by N_t gives

$$A \left(\frac{\hat{K}_t}{\hat{L}_t} \right)^\alpha = A \left(\frac{K_t}{L_t} \right)^\alpha = \frac{w_t}{1-\alpha}, \quad \text{and} \quad A \left(\frac{\hat{K}_t}{\hat{L}_t} \right)^{\alpha-1} = A \left(\frac{K_t}{L_t} \right)^{\alpha-1} = \frac{r_t}{\alpha}. \quad (25)$$

4.4. Pension System

The government collect pension taxes from the labor income and finance the pension payments ℓ and p . Let pension fund balance be B_t . The budget constraint is given by

$$dB_t = \left[r_t B_t + N_t \int \tau_w(w, z) w z d\mathfrak{M}_t - N_t \ell \mathfrak{M}_t(\text{retiring with lumpsum}) - N_t p \mathfrak{M}_t(\text{retired with monthly}) \right] dt. \quad (26)$$

where Z_t is a Brownian motion. Detrending gives

$$d\hat{B}_t = \left[(r_t - \eta_t) \hat{B}_t + \int \tau_w(w, z) w z d\mathfrak{M}_t - \ell \mathfrak{M}(\text{retiring with lumpsum}) - p \mathfrak{M}(\text{retired with monthly}) \right] dt. \quad (27)$$

The pension benefit and the pension taxes stop when the fund balance hits zero. In other words, after $\hat{B}_t$ reaches zero, the τ , ℓ and p are cut to zero for the rest generations.

4.5. Equilibrium

Given the initial distribution $\mathfrak{M}_0(da, ds, dz, dR)$, pension fund balance $\hat{B}_0$, the recursive equilibrium consists of optimal consumption policy path c_t , the optimal retirement policy R_t , the price paths $\{r_t, w_t\}$, the fund balance path $\hat{B}_t$ and the distribution $\mathfrak{M}_t \in \mathcal{P}(\mathbb{R}_+ \times [0, T] \times \mathcal{Z})$ such that

- (a) Given the prices r_t, w_t and $\hat{B}_t$, the optimal consumption policies and the retirement policies solves the individuals optimization problem.
- (b) Given the prices r_t and w_t , the firms optimize.
- (c) The prices are such that the goods market and the labor market clear:

$$\hat{Y}_t = \int c_t d\mathfrak{M} \quad (28)$$

$$\hat{L}_t = \int \exp(\bar{z}(s) + z) \mathbf{1}_{\{\text{working}\}} d\mathfrak{M}. \quad (29)$$

- (d) The individuals optimal retirement decision R_t is consistent with the fund balance path $\hat{B}_t$.
- (e) The distribution solves the corresponding Kolmogorov forward equation.

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